

Annexure1

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. **TITLE: RFID-Based Smart Tagging and Captive Portal System for Efficient Bag and Smartphone Management-** A system that allows for the smooth collecting of bags/mobiles by integrating existing RFID cards and using a captive portal for the pre-registration of devices, avoiding the need for human tagging.

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2. **INVENTOR(S)/ STUDENT(S):**

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3. **DESCRIPTION OF THE INVENTION**

A. PROBLEM ADDRESSED BY THE INVENTION

Applying antiquated techniques to manage the pickup and return of items like electronics or bags during crowded events, like conferences or campus meetings, can be disruptive. To retrieve their stuff, people must fill out paper paperwork, wait in long queues, and carry delicate slips under traditional methods.

Afterwards. This slows down operations, particularly during rush hour, and creates confusion when employees manually match slips to merchandise. In addition to being sluggish, paper slips are unsafe. It could be difficult for someone to get their belongings back if they misplace their slip. A slip could be stolen or copied, allowing someone else to seize another person's possessions. These problems waste time, increase security threats, and irritate guests in addition to adding to the organisers' workload.

To address this, a contemporary approach uses a seamless digital procedure in place of paper slips. Participants register their identities using their university registration number on their smartphones and their university RFID cards. As users wait in queue, they also use a basic captive web interface to list the products they are depositing. They get an identifying tag for their bag, verify their registration, and scan their ID card at check-in. After attaching the tag to their bag, they simply leave their possessions behind and proceed.

Upon entering the lobby, guests scan their ID cards to get their belongings. While waiting in queue, individuals recover their registered items and check out via the captive portal. They scan their ID at the exit gate, and the system instantly confirms the bag tag. They can leave the tag and go without any problems if everything matches.

By using the gateway to track tags and device IDs, the system improves security. The distinctive marking warns employees at checkout if someone tries to depart with the incorrect bag, avoiding theft or confusion. To ensure safety, an alert is sent off at the gate if a mobile device is taken out without the required checks. The procedure is simple to follow; staff members can concentrate on helping people rather than managing paperwork, and participants don't need to learn anything new.

The method also saves trash by doing away with paper slips and forms, which makes it economical and environmentally beneficial for event planners. This strategy maintains efficiency for events with lots of foot traffic, including vendor fairs or student orientations. In addition to organisers avoiding the headache of manual tracking, attendees no longer have to worry about misplacing their slips or standing in queue. There is no need for specialised training or equipment because the usage of ID cards and pre-tagged bags fits in perfectly with current workflows. Even in the most hectic settings, this technology transforms a laborious, prone to mistakes work into a rapid, safe, and effective procedure.

B. STATE OF THE ART/ RESEARCH GAP:

SN	Patent number	Title	Abstract	Difference
1	US20020185532A1	RFID passive data gathering for inventory management	RFID-based system for real-time inventory tracking, error tolerance adjustment, and converting non-	With an emphasis on inventory management, fault tolerance, and data categorisation, US20020185532A1 aims to gather and utilise data from RFID tags linked to goods of interest . It is very different from RFID attendee management that is event-specific.

		and supply chain tracking .	RFID items, enhancing management in various storage facilities.	-vs- Its absence of features designed for event-specific requirements, like smartphone integration, paperless processes, real-time participant verification, and event-driven security warnings, is the main distinction. By integrating dynamic security features like gate-triggered alarms, connecting user identity to products prior to check-in, and customising the system for hectic event settings, the suggested approach directly meets these unmet needs.
2	US20050258961A1	An RFID system for inventory control, item tracking, restocking, and compatibility	an RFID-based inventory system to track item compatibility, refill requirements, and food status. It tracks objects and elapsed time using RFID tags and interrogators. Items that need maintenance or	an RFID-based inventory system to track item compatibility, refill requirements, and food status. It tracks objects and elapsed time using RFID tags and interrogators. Items that need maintenance or replacement are indicated via sensors attached to the tags. US20050258961A1 is designed to use RFID to manage inventory, namely to track food state, refill requirements, and item compatibility [2]. It is not intended for high-throughput event check-in/check-out scenarios, in contrast to the suggested solution. Although it makes use of RFID interrogators, it is devoid of the elements that characterise the suggested event management solution, such as captive portals, paperless workflows, customised attendee-item binding, and attendee self-service. The suggested approach is designed to control event attendance and guarantee a safe and effective movement of people.

		checks	replacement are indicated via sensors attached to the tags.	
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C. DETAILED DESCRIPTION (Technical as well as non-Technical)

In this invention, I offer a comprehensive solution based on Internet of Things (IoT) technology, which includes a captive portal, an ESP32 WiFi module, an RFID reader, buzzers, and a Raspberry Pi that serves as the central controller and data management system.

The system is intended to provide secure and efficient check-in and check-out processes for users and their belongings. The captive portal, which runs on the ESP32 module, handles user registration. When a user registers using their smartphone, they enter their registration ID and belongings list. The ESP32 then grabs the device's MAC address and stores it in the database alongside the registration number. The registration number is the primary key, and the MAC address serves as the secondary key, ensuring unique identification and tracking. In addition, a timestamp is entered to keep track of check-ins in the Raspberry Pi's central database.

Upon arrival, the user scans their RFID card at the reader. The system takes the card's unique registration number and sends it to the central controller for verification. If the registration number is located in the database, the system logs the check-in and updates the user's timestamp. The user's bag is then assigned an RF tag, which is linked to the database's primary key. Once tagged, the bag is safely deposited.

When the user returns to get their possessions, they must first scan their RFID card and check in again. The RFID reader sends the registration number to the controller, which determines whether the current check-in time exceeds the predetermined checkout duration. If the registration number is validated within the timeframe specified, the check-in timestamp is reset to "0" to signify that the user is present and ready to receive their stuff.

As the user approaches the checkout, they must complete the transaction through the captive gateway. The gateway makes a request to the controller, containing the device's registered MAC ID. The controller confirms that the MAC ID exists in the database and double-checks the check-in status. If the check-in timestamp is "0", the controller sets the checkout column for the related primary key to "1".

At the exit checkpoint, the user scans both their RFID card and the allotted RF-tag for their bag to complete the checkout. The RFID controller transmits the data to the main controller, which

confirms that the checkout status is "1" and that the bag-tag matches the stored information. If both conditions are met, the system allows you to exit. If there is a discrepancy, such as an unmatched bag tag or an invalid checkout status, the buzzer sounds to alert the authorities.

The system's main controller manages a number of essential functions aimed at improving security and automation. It maintains a large database that includes user registrations, MAC IDs, timestamps, RF tags, and checkout statuses. To ensure that activities run smoothly, the Python controller program enforces business rules such as check-in and check-out times, user authentication, and error handling. Rule enforcement is critical for assuring compliance with check-in restrictions and checkout times, as well as generating alerts for expired checkouts.

The system also keeps complete data of people who receive warnings, including registration numbers, device IDs, models, check-in times, and belonging lists. Administrators have manual override powers to change entries and exits as needed. The system also has a checkout expiry management function, which lists registration numbers of users whose checkout term has expired and offers choices for extending check-in and checkout periods. This solution reduces security concerns and automates user and possessions tracking via IoT-based automation and intelligent data management, resulting in a streamlined and error-free operation.

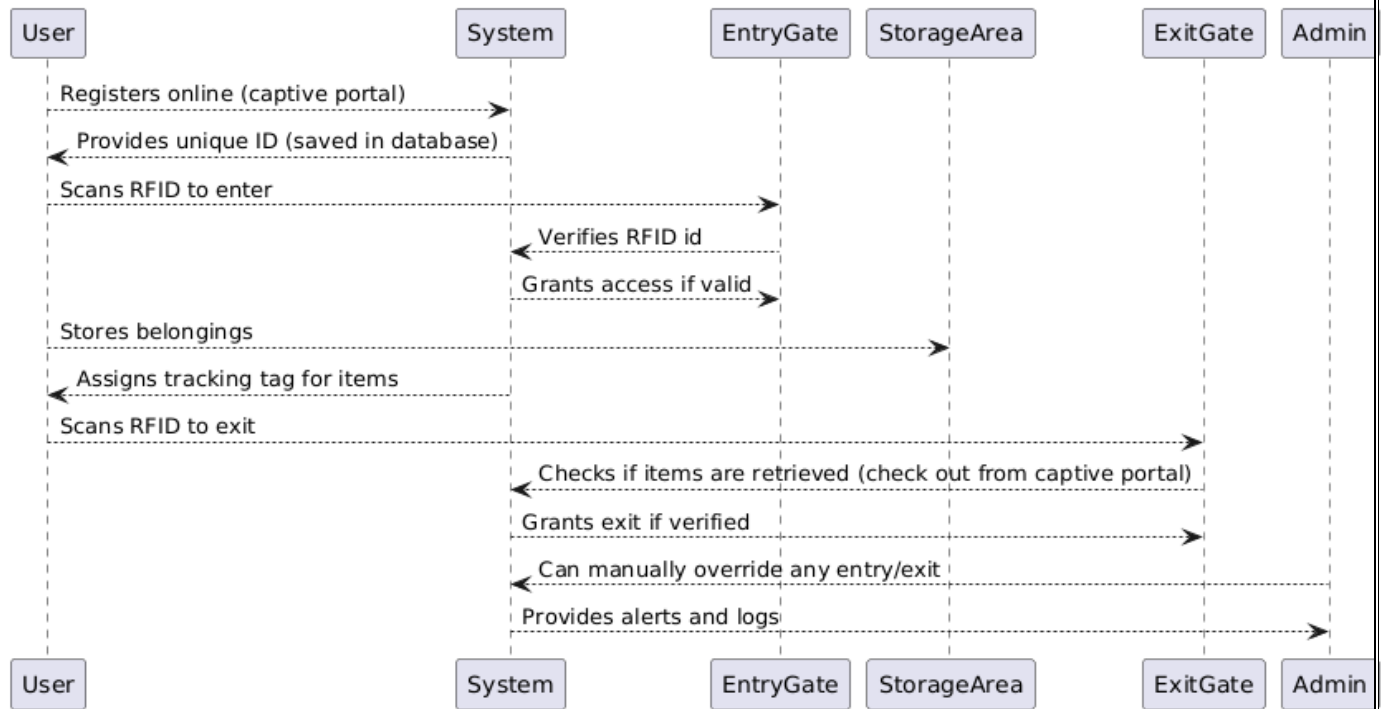


Figure 1. system process from regiteration to checkout

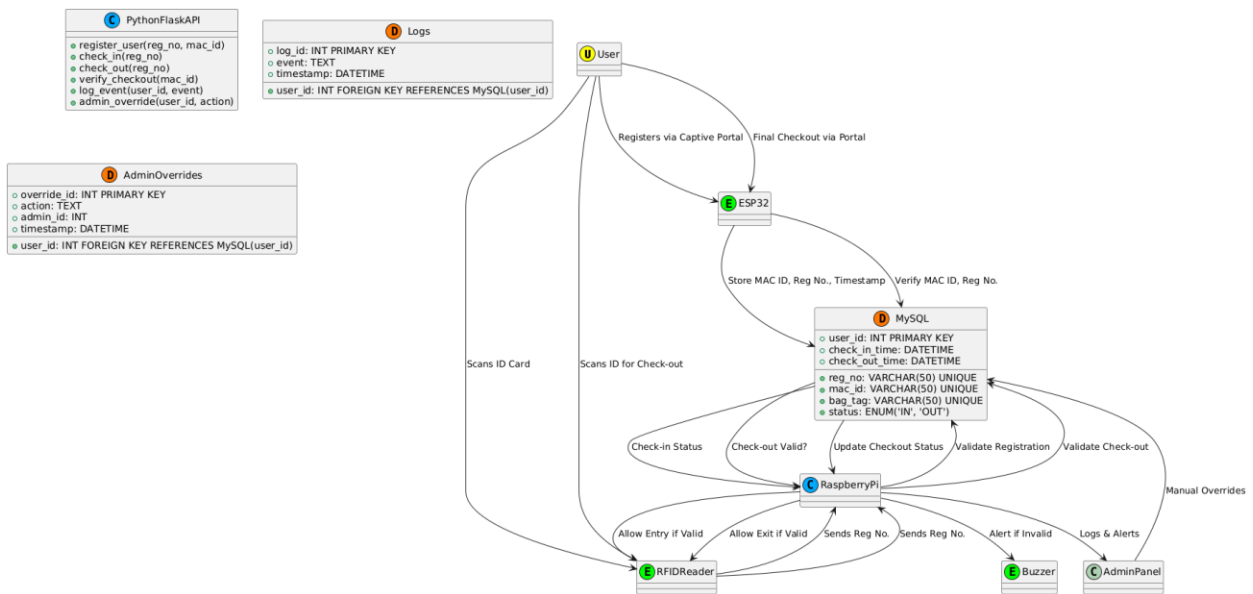


Figure 2. technical detailed how whole system work

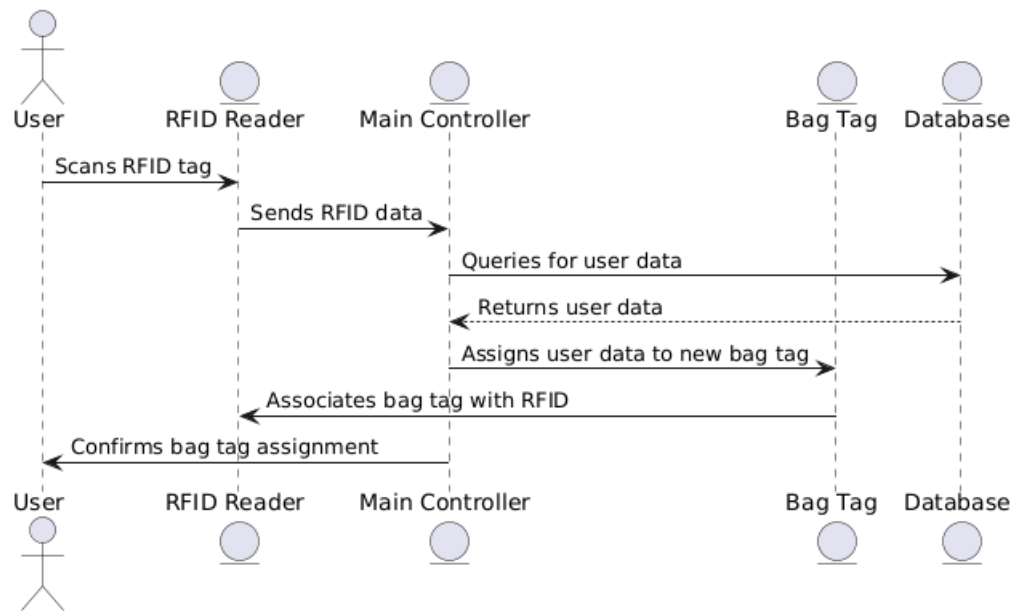


Figure 3. how bag-tag assign when a user check in

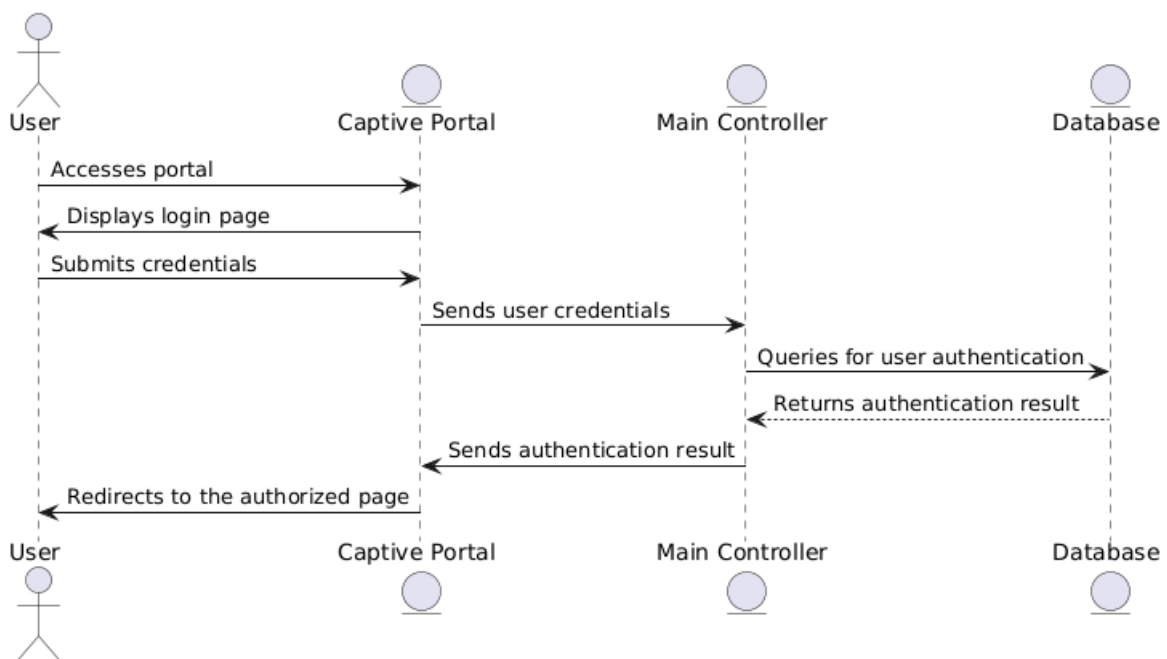


Figure 4 . how Registration process done by using captive portal

D. ADVANTAGES OF THE INVENTION:

- a. **Enhanced Efficiency:** By replacing manual paper-based processes with a digital workflow, the system significantly reduces check-in and checkout times, allowing attendees to deposit and retrieve their belongings quickly, even during high-traffic events like conferences or student orientations.
- b. **Improved Security:** The integration of RFID tags and MAC address verification ensures that only authorized users can retrieve their items, minimizing the risk of theft or mix-ups. Alerts triggered by mismatched tags or unauthorized checkouts further bolster security.
- c. **Paperless Operation:** Eliminating the need for paper slips reduces administrative overhead, cuts costs for organizers, and promotes an eco-friendly approach by minimizing waste, aligning with sustainability goals.
- d. **Seamless User Experience:** Attendees use familiar tools like their university RFID cards and smartphones, requiring no additional training. The intuitive captive portal simplifies registration and checkout, enhancing convenience.
- e. **Real-Time Tracking:** The system provides organizers with real-time visibility into item deposits and retrievals, enabling better crowd management and resource allocation during busy events.
- f. **Scalability:** Designed to handle large volumes of users, the system adapts effortlessly to events of varying sizes, from small gatherings to large-scale vendor fairs, without compromising performance.
- g. **Cost-Effectiveness:** By leveraging existing RFID infrastructure and affordable IoT components like ESP32 and Raspberry Pi, the system offers a low-cost solution that delivers high value without requiring expensive new equipment.
- h. **Error Reduction:** Automation of tag assignment, identity verification, and checkout processes minimizes human errors, such as misplaced slips or incorrect item matching, ensuring a reliable operation.
- i. **Flexibility for Administrators:** The inclusion of manual override options and checkout expiry management allows staff to adapt the system to unexpected situations, maintaining operational control.
- j. **Data-Driven Insights:** Detailed logs of user activity, including timestamps and alert records, provide valuable data for post-event analysis, helping organizers improve future events and address security incidents effectively.

KEYWORDS:

- a. RFID Tagging
- b. Captive Portal
- c. IoT Automation
- d. Smart Management
- e. Security Alerts
- f. Paperless Workflow
- g. Real-Time Tracking
- h. Smartphone Integration
- i. Event Efficiency
- j. Scalable System

POTENTIAL CLAIMS:

1. A system for managing bags and smartphones at events, comprising an RFID reader, a captive portal hosted on an ESP32 module, and a Raspberry Pi controller, wherein user identity is linked to deposited items via RFID tags and verified through a smartphone-registered MAC address.
2. The system of Claim 1, wherein the captive portal enables pre-registration of user belongings using a university registration number, eliminating manual tagging and paper slips.
3. The system of Claim 1, further comprising a database managed by the Raspberry Pi that stores registration numbers, MAC addresses, timestamps, and RF-tag assignments for real-time tracking and verification.
4. A method for secure check-in, wherein a user scans an RFID card, the system assigns an RF-tag to their bag, and the tag is linked to the user's registration number in the database, ensuring accurate item association.
5. A method for secure checkout, wherein the system verifies the user's RFID card and bag tag at an exit gate, triggering a buzzer alert if the tag or checkout status does not match the database record.
6. The system of Claim 1, wherein IoT-based automation enforces predefined check-in and checkout periods, with alerts generated for expired checkouts to enhance security and compliance.
7. The system of Claim 1, further comprising an administrative override feature, allowing manual adjustments to check-in/check-out statuses and extension of expiry periods, providing operational flexibility.

Signature of Inventor

Name of Inventor with UID and
Department

Date